

4.11.4 Data Summarization & Statistical Analysis

گروپ بائی ایک جیسے ریکارڈ جوڑتا ہے، پھر COUNT, SUM, AVG, MAX, MIN سے خلاصہ نکلتا ہے۔

CS XI

Chapter 4: Data and Analysis

Lecture 14 of 15

4.11.4 Data Summarization & Statistical Analysis

LEARN IT

Grouping combines rows that share a value (all CS students together) and then an **aggregate** calculates a number for the group. In Access click Totals (Σ) to reveal the Total row in the grid.

Function	What it calculates
COUNT()	How many records (students in IT)
SUM()	Total of a number field (fees collected)
AVG()	Mean
MAX() / MIN()	Highest / lowest in the group

WORKED EXAMPLE

Average marks per department: add Department and Marks; Totals Σ ; Department → Group By; Marks → Avg; Run.

Department	Avg of Marks
Computer Science	82.5
IT	76.0

Check: CS marks 80, 85 → average $(80+85)/2 = 82.5$. IT 70, 82 → 76.0.

```
SELECT Department, AVG(Marks) AS AvgOfMarks
FROM Student
GROUP BY Department;
```

اردو میں سمجھیں

گروپ بائی ایک جیسے ریکارڈ جوڑتا ہے، پھر COUNT, SUM, AVG, MAX, MIN سے خلاصہ نکلتا ہے۔ ایکسیس میں Σ بٹن Total قطار کھولتا ہے۔

CHECK YOURSELF

1. Which Total setting goes under Department?
2. Which function counts how many books each member has borrowed?

Answers: 1. Group By. 2. COUNT (of BorrowID or BookID) grouped by MemberID.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Grouping in a query:

- a. Collapses rows that share a value (e.g. by class)
- b. Deletes the table
- c. Draws OSI
- d. Opens Figma

MCQ 2. Count(*) with group by class gives:

- a. How many students per class
- b. The PK
- c. A pie automatically
- d. A stack

MCQ 3. Avg(Marks) is:

- a. An aggregate
- b. A primary key
- c. A form control always
- d. A foreign key

MCQ 4. Sum is for:

- a. Totals (fees, marks)
- b. Sorting names alphabetically
- c. Encryption
- d. Routing

MCQ 5. Min/Max in a totals query:

- a. Smallest/largest in the group
- b. The table name
- c. The form wizard
- d. TCP ports

MCQ 6. You summarise when:

- a. n is large and you need a picture of each group
- b. You have one row
- c. You draw NAND
- d. You write sprints

Extra short questions

1. Name four aggregate functions.
2. What must you set in Access Totals query?
3. Example: average marks per subject.
4. Why not list 2000 rows to the principal?

Extra long question

1. From a Marks table, design two totals queries: (1) count of fails per class (2) average marks per subject. Show the Group By row.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Count, Sum, Avg, Min, Max.
2. Group By on the category; Count/Avg on the measure.
3. Group by Subject; Avg(Marks).
4. A 5-row summary is decidable; a dump is not.

Long

1. (1) Class Group By; Marks Count with criteria <40 or a Fail Yes/No Count. (2) Subject Group By; Marks Avg. Sample output.